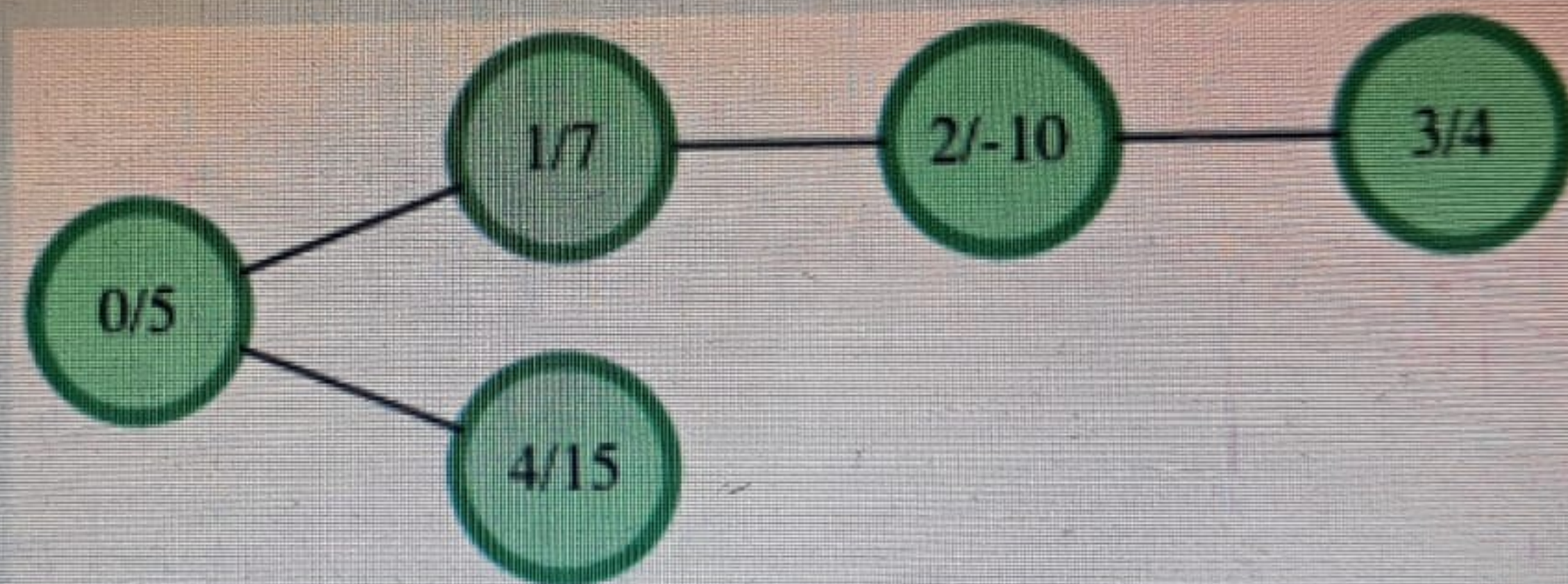


Given a tree rooted at node 0 and a *value* assigned to each node, determine the maximum sum of the values along any path in the tree. The path must not be empty, and in some cases it might not go through the root. The following tree (labeled *node number/value*) is analyzed below.



Two of the possible paths are $4 \rightarrow 0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ which has a sum of $15 + 5 + 7 + -10 + 4 = 21$ and $1 \rightarrow 2 \rightarrow 3$ with a sum of $7 + -10 + 4 = 1$. A third possible path, the maximum sum path, is $4 \rightarrow 0 \rightarrow 1$ with a sum of $15 + 5 + 7 = 27$.

Function Description

Function Description

Complete the function *bestSumAnyTreePath* in the editor below. It must return an integer that represents the largest value sum on any non-empty path in the tree.

bestSumAnyTreePath has the following parameter(s):

- *parent* *int*[*n*]: integer array where *parent*[*i*] = *j* means that node *j* is a parent of node *i*. *parent*[0] will be set to -1 to indicate that node 0 is the root of the tree.
- *values* *int*[*n*]: integer array where *values*[*i*] denotes the value of node *i*

Constraints

- $1 \leq n \leq 10^5$
- *parent*[0] = -1
- $0 \leq \text{parent}[i] < n$ for $1 \leq i < n$
- $-1000 \leq \text{values}[i] \leq 1000$

```
10  /*
11  *
12  *
13  *
14  *
15  *
16  *
17  *
18
19  in
20
21  }
22
23  > in
```


Constraints

- $1 \leq n \leq 10^5$
- $\text{parent}[0] = -1$
- $0 \leq \text{parent}[i] < n$ for $1 \leq i < n$
- $-1000 \leq \text{values}[i] \leq 1000$
- the parent array defines a valid tree

► Input Format Format for Custom Testing

▼ Sample Case 0

Sample Input

5
-1
0
1
2
0
5
-2
10

5
-1
0
1
2
0
5
-2
10
10
-3
10

Sample Output

28

Explanation

The tree looks like:



Language

1 >
9
10 /*
11 *
12 *
13 *
14 *
15 *
16 *
17 */
18
19 in
20
21 }
22
23 > in



27. What Comes Next? 4

ALL

Predict the figure in (?) place.



24

25

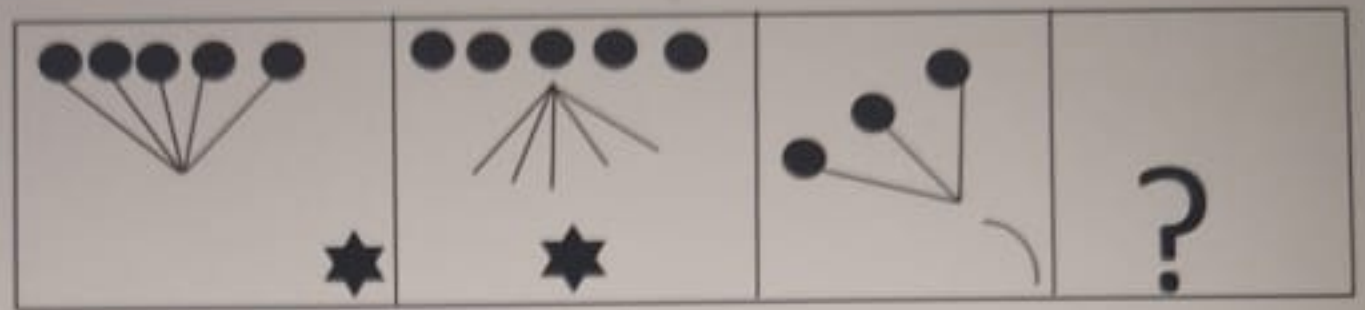
26

27

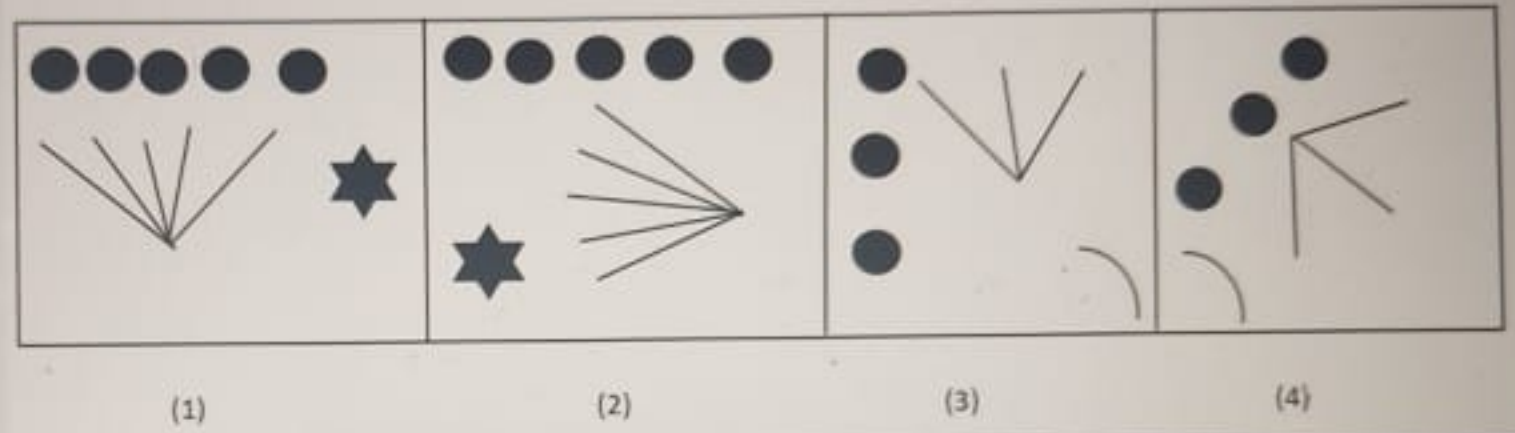
28

29

30



Answer Fig:



ALL



24

25

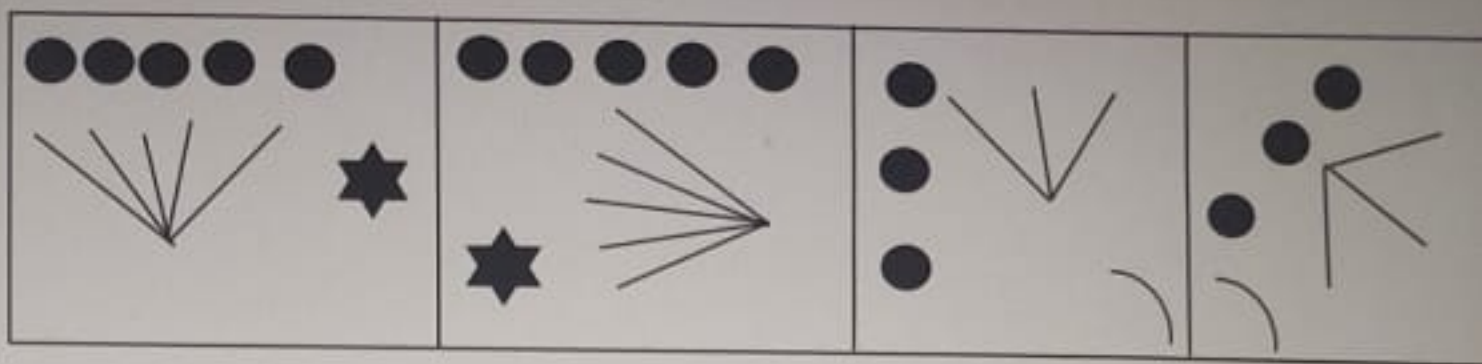
26

27

28

29

30



(1)

(2)

(3)

(4)

Pick **ONE** option

☐ 1

☐ 2

☐ 3

☐ 4

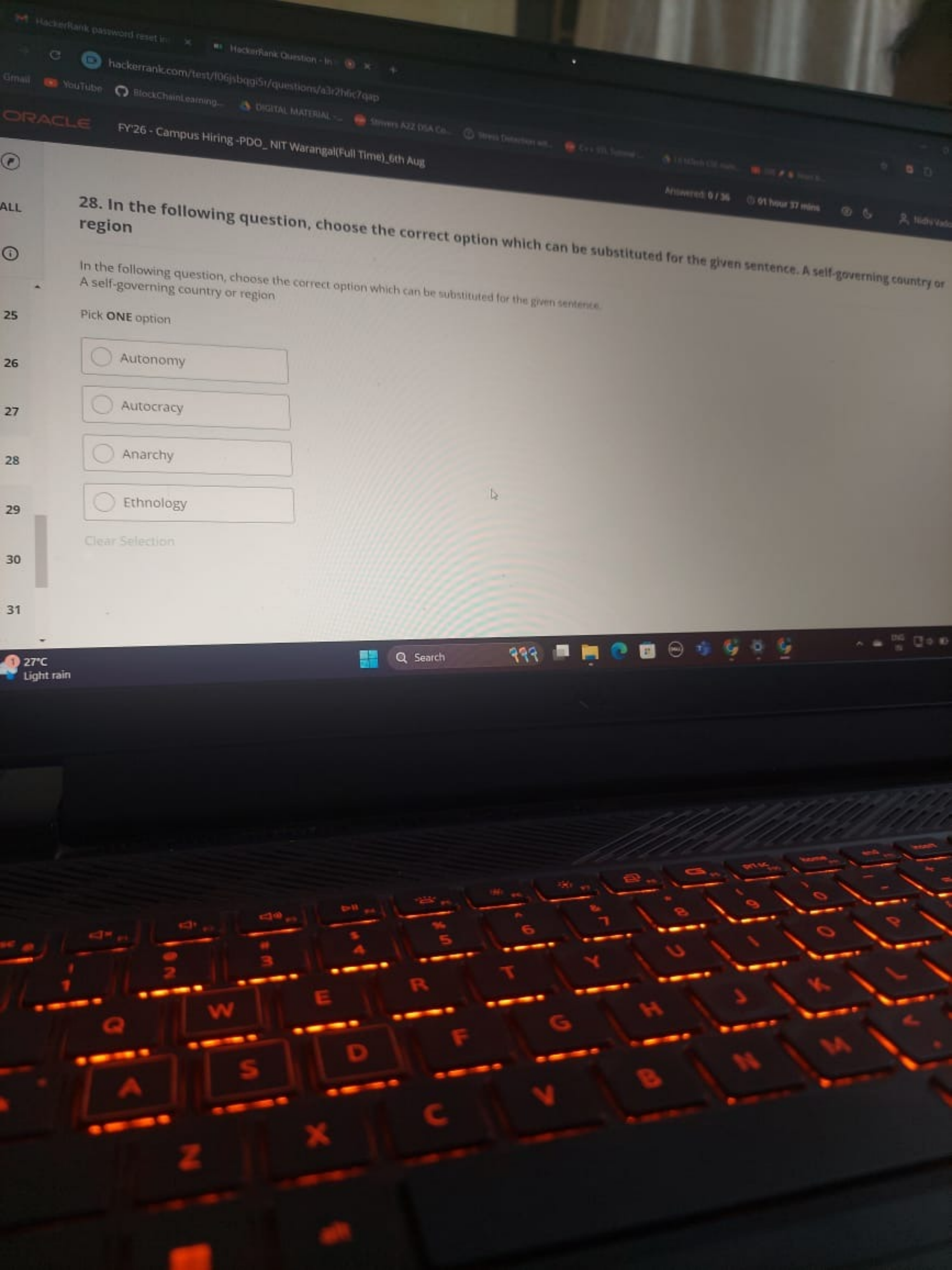
Clear Selection

27°C
Light rain



Search





28. In the following question, choose the correct option which can be substituted for the given sentence. A self-governing country or region

In the following question, choose the correct option which can be substituted for the given sentence.
A self-governing country or region

Pick **ONE** option

☐ Autonomy

☐ Autocracy

☐ Anarchy

☐ Ethnology

Clear Selection



ALL



29. I don't think we should hire him. He seems like _____.

I don't think we should hire him. He seems like _____

Pick **ONE** option

26

☐ A fish out of water

27

☐ The black sheep of the family

28

☐ A wolf in sheep's clothing

29

☐ The apple of my eye

30

Clear Selection

31

32

<https://www.hackerrank.com/test/106jsbqgi5r/questions/90b47chdbas>



Search



tab

caps lock

shift



ALL



30. I've been studying for this exam for months, so I'm hoping to _____

I've been studying for this exam for months, so I'm hoping to _____

Pick **ONE** option

27

☐ Hit the ground running

28

☐ Have a field day

29

☐ Be a dark horse

30

☐ Have a lot on my plate

Clear Selection

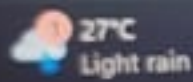
31



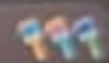
32

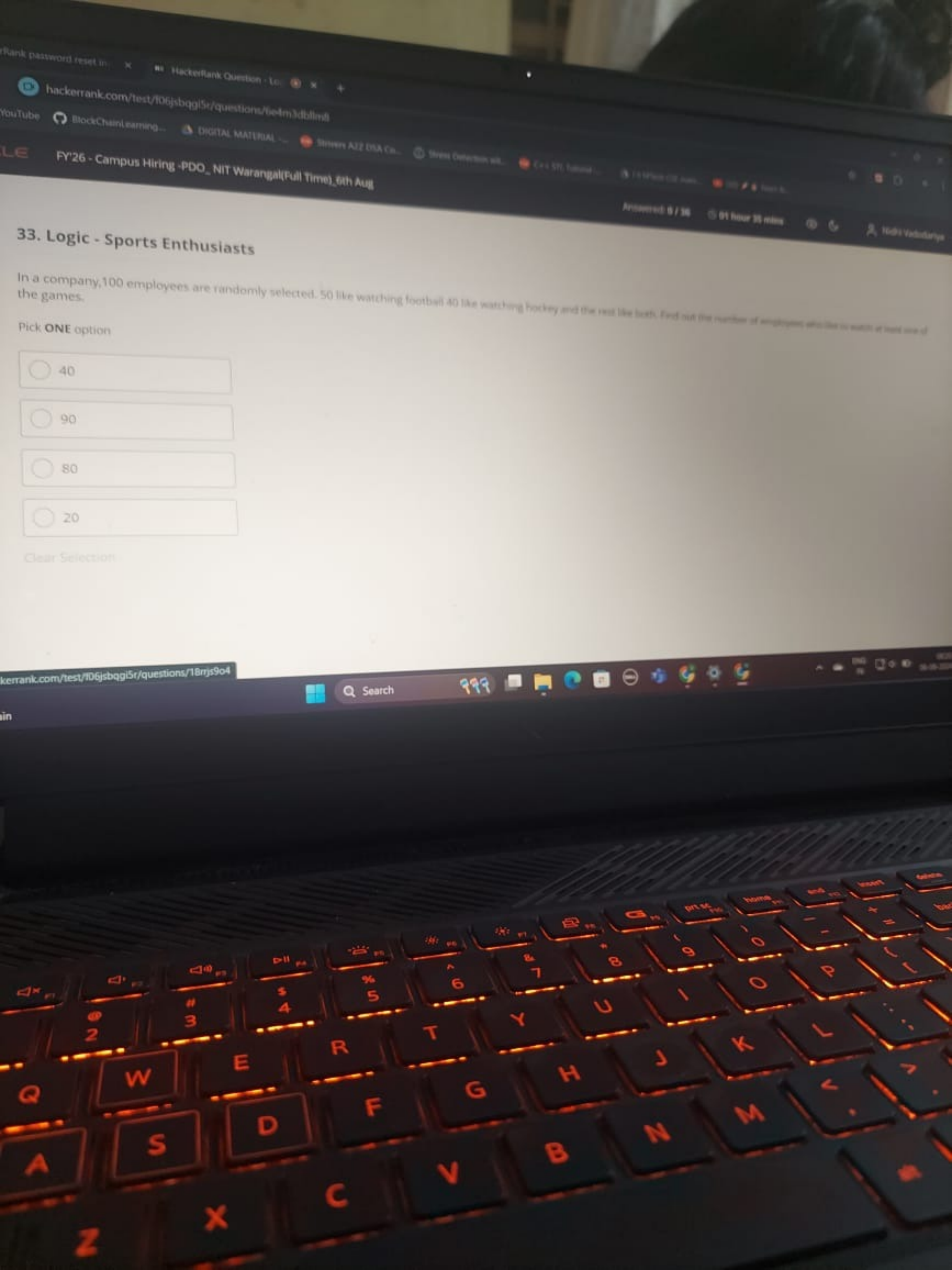
33

<https://www.hackerrank.com/test/106jsbqgi5r/questions/dgfobh0dr9n>



Search





33. Logic - Sports Enthusiasts

In a company, 100 employees are randomly selected. 50 like watching football 40 like watching hockey and the rest like both. Find out the number of employees who like to watch at least one of the games.

Pick **ONE** option

- ☐ 40
- ☐ 90
- ☐ 80
- ☐ 20

[Clear Selection](#)

ALL

34. Integer Division

What is the remainder of 1992 divided by 92?

Pick **ONE** option

☐ 0

☐ 1

☐ 40

☐ None of the above

Clear Selection

34

35

36

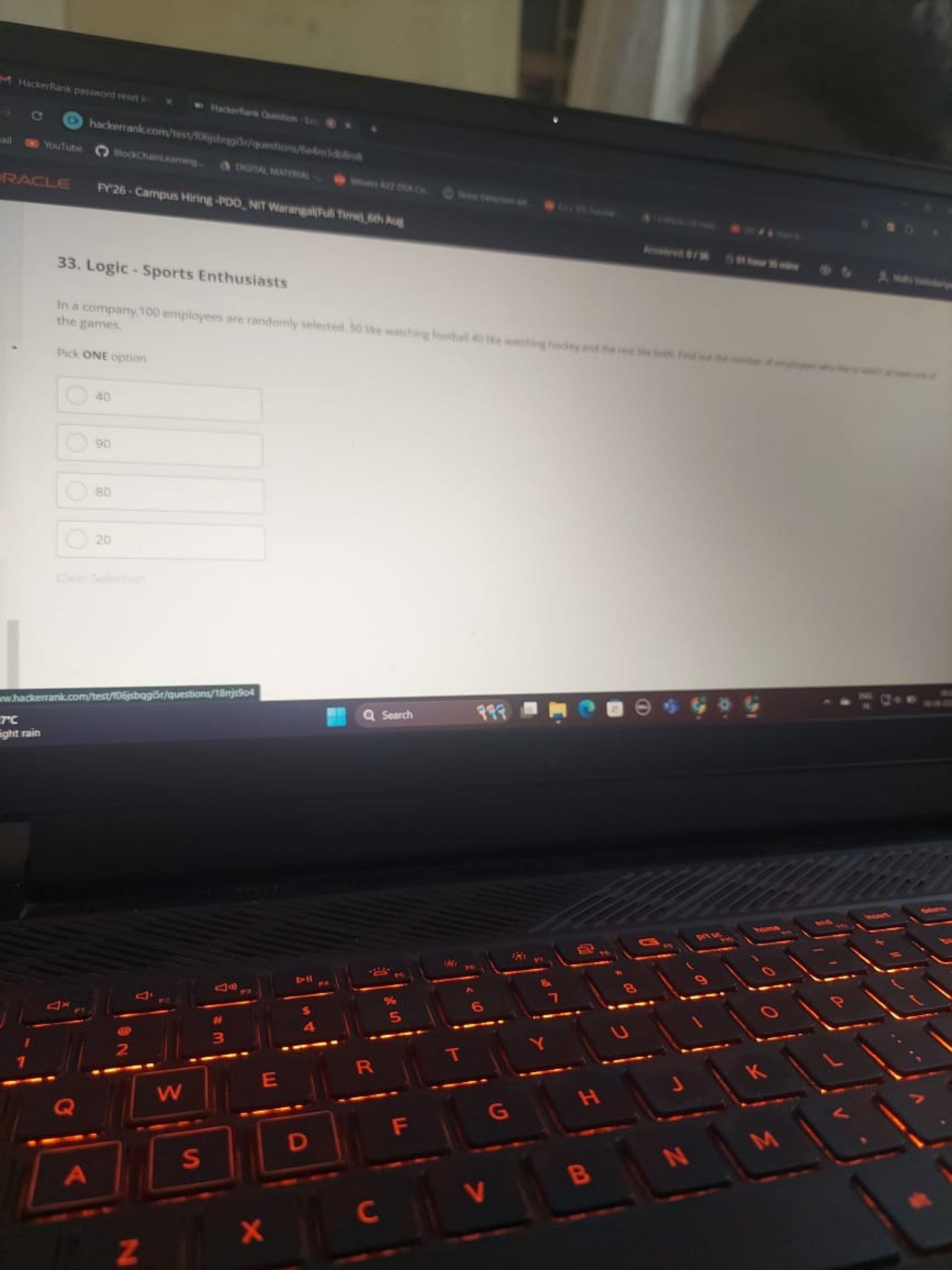
<https://www.hackerrank.com/test/106jsbqgi5r/questions/18rjs9o4>

27°C
Light rain



Search





33. Logic - Sports Enthusiasts

In a company, 100 employees are randomly selected. 50 like watching football 40 like watching hockey and the rest like both. Find out the number of employees who like to watch at least one of the games.

Pick **ONE** option

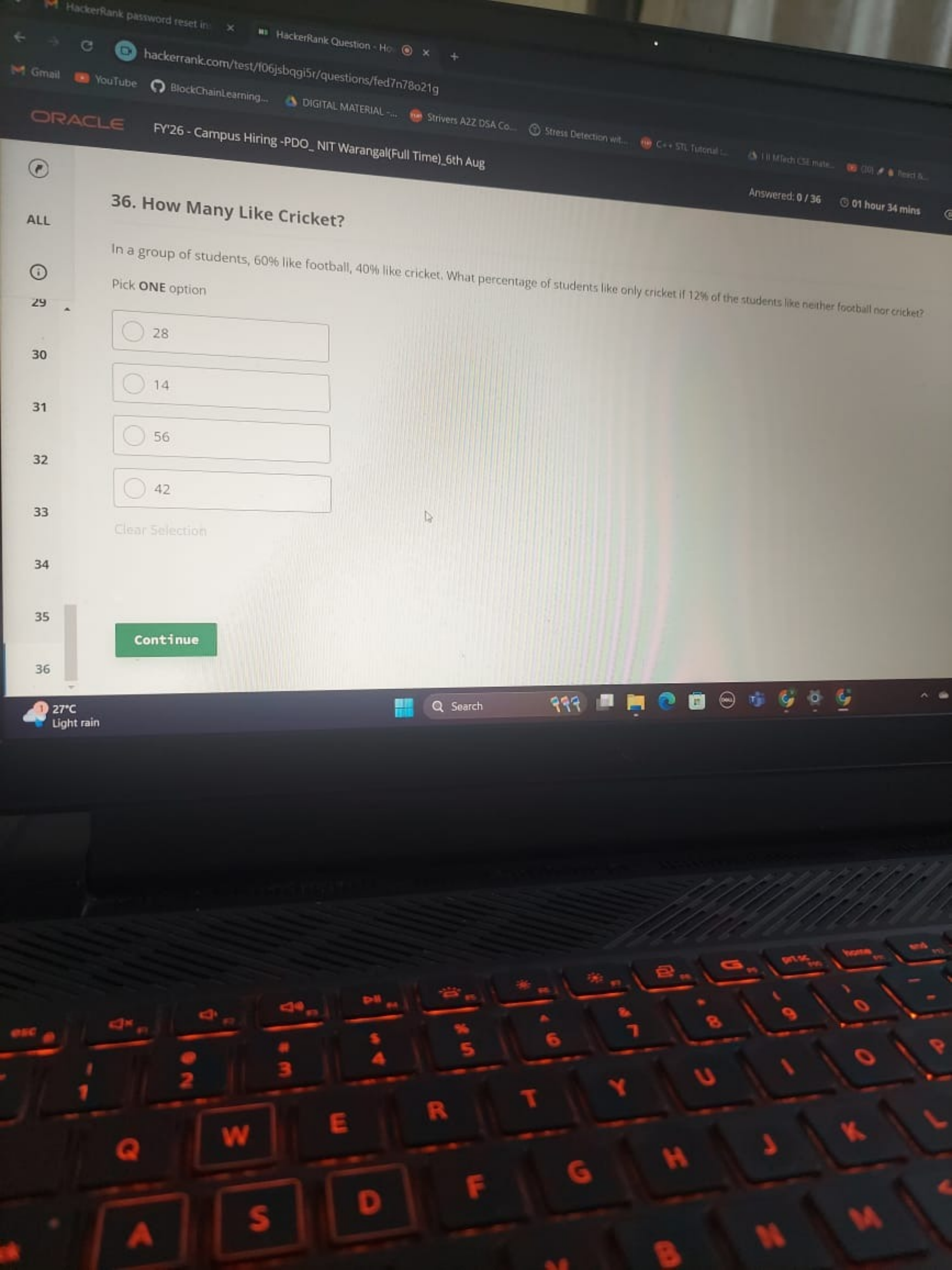
☐ 40

☐ 90

☐ 80

☐ 20

[Clear Selection](#)



36. How Many Like Cricket?

In a group of students, 60% like football, 40% like cricket. What percentage of students like only cricket if 12% of the students like neither football nor cricket?

Pick **ONE** option

- ☐ 28
- ☐ 14
- ☐ 56
- ☐ 42

Clear Selection

Continue

27°C
Light rain



Search

